

Project #1: Incorporating Economic Data

ECON400 | Carolina Varanko

Introduction

This project uses annual U.S. data on Gross Domestic Product (GDP) and Personal Consumption Expenditure (PCE) from the Bureau of Economic Analysis (BEA), from the 1970s to 2025, to quantify the relationship using Ordinary Least Squares (OLS) regression. Two model specifications are estimated and compared: Model A using dollar values, and Model B using a log-log transformation.

Models and Methods

The analysis uses Ordinary Least Squares (OLS) regression to analyze the relationship between GDP and PCE. Two specifications are estimated: Model A regresses PCE directly on GDP in dollar values, while Model B applies a natural log transformation to both variables (log-log model). Both models include a constant intercept.

Model A: $PCE = \beta_0 + \beta_1(GDP) + \varepsilon$

Model B: $\ln(PCE) = \beta_0 + \beta_1 * \ln(GDP) + \varepsilon$

Each model had a t-test conducted on the estimated coefficients at the 95% confidence level to assess statistical significance. The overall fit of each model is evaluated using R-squared and Adjusted R-squared.

Data Description

Annual data on GDP (Line 1) and PCE (Line 2) were taken from the BEA Table 1.1.5 (GDP and Personal Income). The dataset contains 56 annual observations from 1970 to 2025. All values are reported in the billions of U.S. dollars. The data was downloaded from the BEA.gov site and cleaned in Excel prior to analysis in R.

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Analysis of Results

Model A:

	Estimate	Standard Error	t-value/p-value
Intercept	-206.80	19.12	-10.81/ <0.001***
GDP (slope)	0.6861	0.001432	479.23 / <0.001***
R-squared	0.9998		
Adj. R-squared	0.9998		

Model A estimates $PCE = -206.80 + 0.6861(GDP)$. The slope tells us that every additional billion dollars in GDP, PCE increases by ~\$0.69 billion. Both coefficients are statistically significant at the 95% confidence level, and R-squared of 0.9998 shows that the GDP explains nearly all the variation in PCE.

Model B:

	Estimate	Standard Error	t-value/p-value
Intercept	-0.8482	0.01638	-51.77/ <0.001***
GDP (slope)	1.0464	0.001826	573.01 / <0.001***
R-squared	0.9998		
Adj. R-squared	0.9998		

Model B estimates $\ln(PCE) = -0.848 + 1.046 * \ln(GDP)$. The slope tells us that a 1% increase in GDP, PCE increases with a ~1.046%, which suggests that consumption grows slightly faster than income over time. Both coefficients are statistically significant, and R-squared remains 0.9998.

Comparing Model's A & B:

	Model A	Model B
R-squared	0.9998	0.9998
Adj. R-squared	0.9998	0.9998
Slope interception	\$0.686 billion per \$1 billion GDP	1.046% per 1% GDP

Both models have the same R-squared, but Model B is the preferred choice. The log-log model is better suited for data that grows rapidly over several decades.

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Conclusion

Overall, both models confirm that as national income grows, consumer spending grows as well. Model A shows that every additional billion dollars in GDP, PCE increases by ~\$0.69 billion, and Model B shows that a 1% increase in GDP leads to ~1.046% increase in PCE. Both models are statistically significant and explain nearly all the variation in PCE with an E-squared of 0.9998. Model B is the preferred model because it is better suited for data that grows rapidly over several decades.

References

U.S. Bureau of Economic Analysis (BEA). (2025). Table 1.1.5: Gross Domestic Product. National Income and Product Accounts. Retrieved from <https://www.bea.gov>

Attachments

R Model Outputs

```
> # ---- MODEL A ----
> modelA <- lm(PCE ~ GDP, data = data)
> summary(modelA)

Call:
lm(formula = PCE ~ GDP, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-228.343  -59.690   -0.125   56.199  202.042

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.068e+02  1.912e+01  -10.81 3.98e-15 ***
GDP           6.861e-01  1.432e-03   479.23 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 85.41 on 54 degrees of freedom
Multiple R-squared:  0.9998,    Adjusted R-squared:  0.9998
F-statistic: 2.297e+05 on 1 and 54 DF,  p-value: < 2.2e-16
```

```
> # ---- MODEL B ----
> data$ln_GDP <- log(data$GDP)
> data$ln_PCE <- log(data$PCE)
>
> modelB <- lm(ln_PCE ~ ln_GDP, data = data)
> summary(modelB)

Call:
lm(formula = ln_PCE ~ ln_GDP, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.032095 -0.007796  0.000170  0.008361  0.023147

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.848202  0.016384  -51.77 <2e-16 ***
ln_GDP       1.046395  0.001826   573.01 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01269 on 54 degrees of freedom
Multiple R-squared:  0.9998,    Adjusted R-squared:  0.9998
F-statistic: 3.283e+05 on 1 and 54 DF,  p-value: < 2.2e-16
```